

# Sonatype Nexus for Artifact Storage

Setting up a private, governed home for every binary your organization produces or consumes

## IN THIS DOCUMENT

- What Nexus Repository Manager is
- Proxy, hosted & group repositories
- Release vs. snapshot repositories
- Governance & security (IQ / Firewall)
- Typical repository layout

Prepared for internal training · Topic 2 of 3

## 1 What Is Nexus?

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Sonatype Nexus Repository Manager is a **binary / artifact repository manager** — effectively a private, controlled version of Maven Central, npm, or Docker Hub, run inside your own organization.

**Why teams use it:** it gives you a single source of truth for every build output — JARs, npm packages, Docker images, .NET packages, and more — with full control over what enters your build pipeline.

## 2 Repository Types in Nexus

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### Proxy Repositories

A proxy repository caches a public repository (Maven Central, npmjs, etc.) locally. This gives you:

- Faster builds — dependencies are served from a local cache instead of the public internet
- Resilience — builds keep working even if the public repository has an outage
- Control — visibility and gatekeeping over exactly what external components are allowed in

### Hosted Repositories

This is where your own organization's built artifacts get published — internal libraries, release builds, and shared components that other teams depend on.

### Group Repositories

A group repository aggregates proxy and hosted repositories behind a **single URL**, so developers and CI servers only need one repository endpoint configured, regardless of where the artifact actually lives.

### Release vs. Snapshot Repositories

| Type     | Mutability                    | Typical Use                        |
|----------|-------------------------------|------------------------------------|
| Snapshot | Mutable — can be overwritten  | In-progress / nightly builds       |
| Release  | Immutable — never overwritten | Versioned, production-ready builds |

### 3 Typical Repository Structure

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```
maven-public (group)
├─ maven-releases (hosted)
├─ maven-snapshots (hosted)
└─ maven-central (proxy)
```

Developers and CI jobs point at `maven-public`, and Nexus internally decides whether to serve a request from its cache, proxy it out to the public internet, or serve it from a hosted internal repository.

### 4 Governance & Security

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Beyond storage, Nexus (particularly with **IQ Server / Firewall** add-ons) provides policy enforcement:

- Quarantining risky or vulnerable components before they reach developers
- Blocking components with disallowed licenses
- Flagging known-vulnerable versions automatically, before they're pulled into a build

**Takeaway:** Nexus isn't just storage — it's the control point between the public open-source ecosystem and everything that actually gets built and shipped inside your organization.

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*Part 2 of 3 — Sonatype Nexus for Artifact Storage. Continued in Part 3: CI Integration with Artifact Repositories.*